

MSDM 5003 Lecture 9

30 October 2025

Unsupervised Learning and PCA

Outline:

1. Unsupervised learning

Principal component analysis (PCA)

2. Maximum variance formulation

3. Minimum error formulation

4. Applications of PCA

- Data compression
- Data pre-processing
- Data visualization

5. Application to portfolio management in finance

Unsupervised Learning¹

So far, we have considered supervised learning, that is, learning from **labeled data**.

What can we get from **unlabeled data**?

Common unsupervised learning approaches:

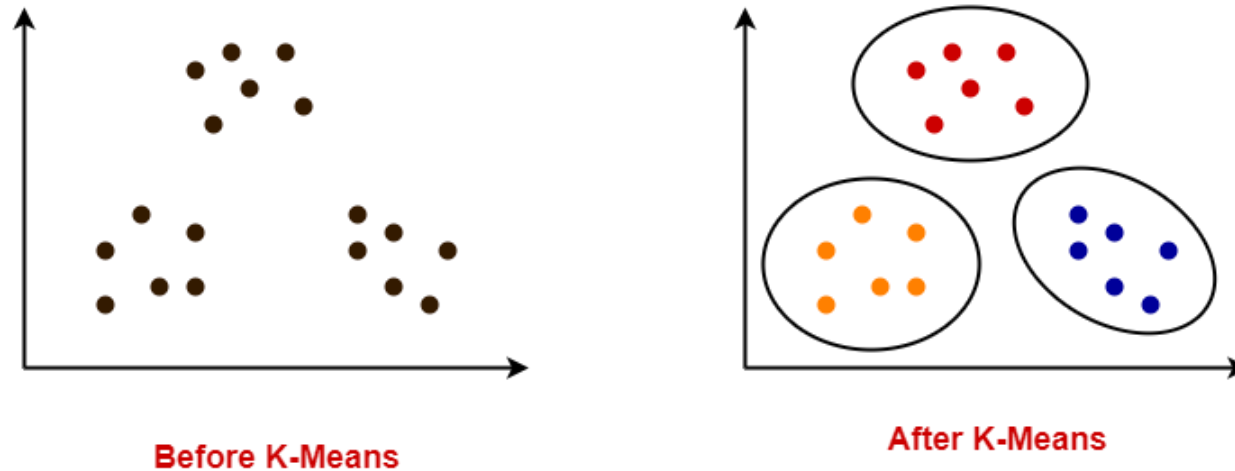
1. Clustering
2. Association rules
3. Dimensionality reduction

¹”What is unsupervised learning?” <https://www.ibm.com/think/topics/unsupervised-learning> (2021).

1.1 Clustering (1)

Clustering: group unlabeled data based on their similarities or differences

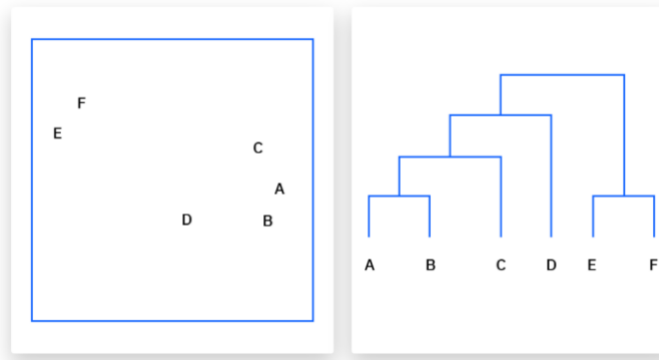
a. Exclusive clustering, e.g. K-means clustering



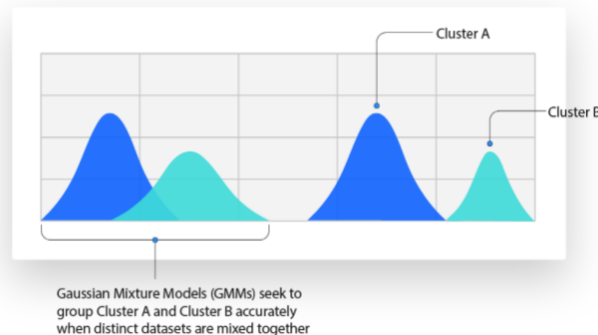
b. Overlapping clustering: allows data points to belong to multiple clusters (soft/fuzzy clustering)

1.1 Clustering (2)

c. Hierarchical clustering: e.g. dendrogram



d. Probabilistic clustering: e.g. Gaussian mixture models



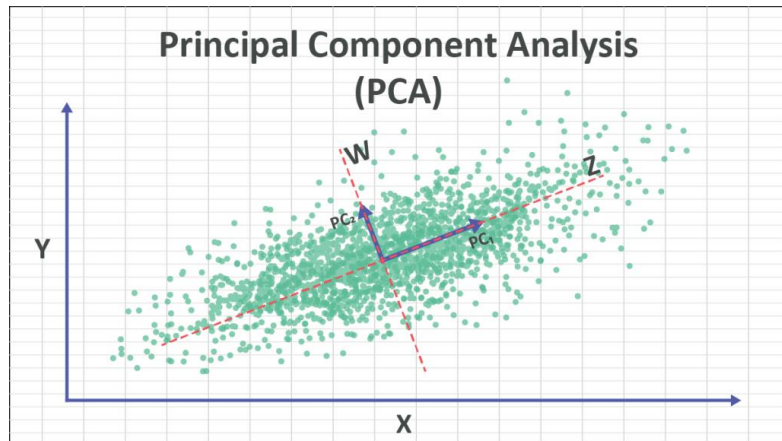
1.2 Association Rules

Recommendation engines, e.g. “Customers who bought this item also bought ...”

Apriori algorithms (rule-based learning of databases)

1.3 Dimensionality Reduction

a. Principal component analysis (PCA)



b. Singular value decomposition
 $A = USV^T$ (SVD)

■ $A = U \Sigma V^T$ - example: Users to Movies

	Matrix	Alien	Serenity	Casablanca	Amelie
1	1	1	0	0	
3	3	3	0	0	
4	4	4	0	0	
5	5	5	0	0	
0	2	0	4	4	
0	0	0	5	5	
0	1	0	2	2	

SciFi-concept Romance-concept

0.13	0.02	-0.01
0.41	0.07	-0.03
0.55	0.09	-0.04
0.68	0.11	-0.05
0.15	-0.59	0.65
0.07	-0.73	-0.67
0.07	-0.29	0.32

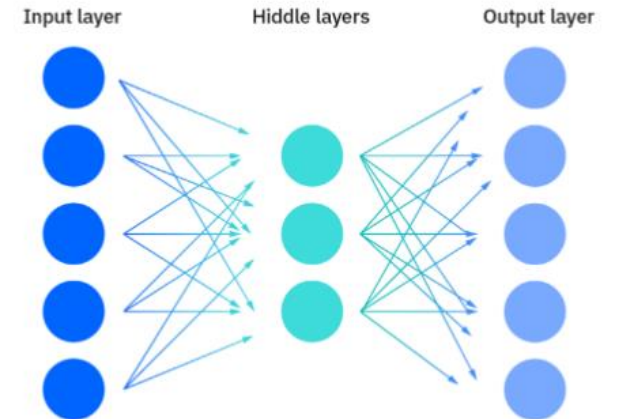
 $\times$

12.4	0	0
0	9.5	0
0	0	1.3

 $\times$

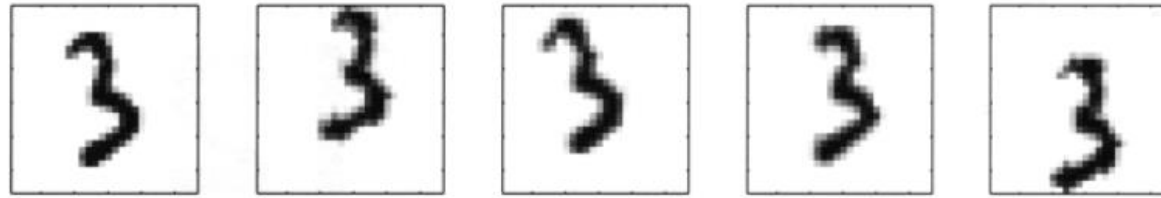
0.56	0.59	0.56	0.09	0.09
0.12	-0.02	0.12	-0.69	-0.69
0.40	-0.80	0.40	0.09	0.09

c. Autoencoders



2. Principal Component Analysis²

In many information processing tasks, a small number of *degrees of freedom* is embedded in a high dimensional space.



A synthetic data set obtained by taking one of the off-line digit images and creating multiple copies in each of which the digit has undergone a random displacement and rotation within some larger image field. The resulting images each have $100 \times 100 = 10,000$ pixels.

In the above example, the degrees of freedom correspond to the horizontal displacement, vertical displacement, and the rotations, although the image has 10^4 pixels.

PCA is a common technique to **extract low dimension features from high-dimensional data**.

There are two formulations of PCA.

² Bishop CM (2006) *Pattern Recognition and Machine Learning*. Springer. Chapter 12.

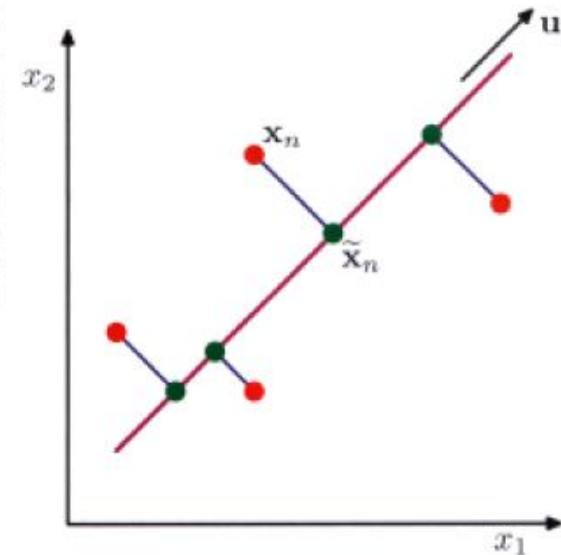
2.1 Maximum Variance Formulation

2.1 The Covariance Matrix

Consider a data set of observations $\{\mathbf{x}_n\}$ where $n = 1, \dots, N$, and $\mathbf{x}_n$ is a D -dimensional variable.

Goal: **Project** the data onto an M -dimensional space with $M < D$ while maximizing the variance of the projected data.

Principal component analysis seeks a space of lower dimensionality, known as the principal subspace and denoted by the magenta line, such that the orthogonal projection of the data points (red dots) onto this subspace maximizes the variance of the projected points (green dots). An alternative definition of PCA is based on minimizing the sum-of-squares of the projection errors, indicated by the blue lines.



The Projection

To begin with, consider the projection onto a one-dimensional space ($M = 1$).

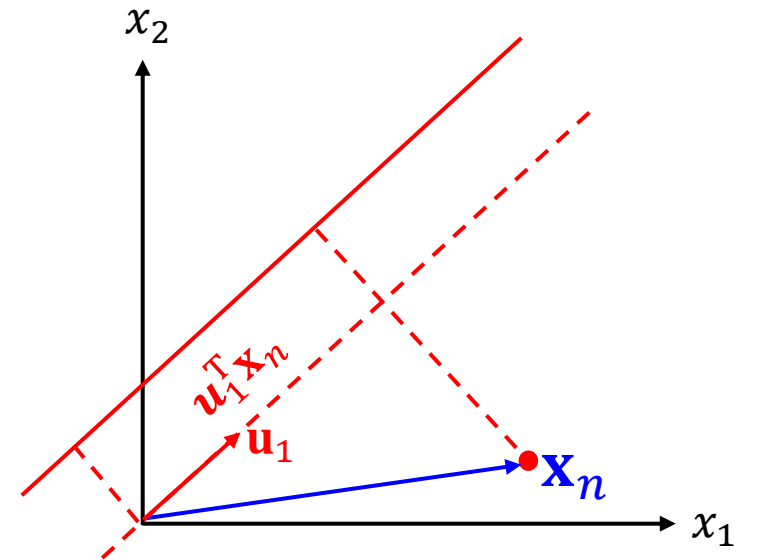
Let $\mathbf{u}_1$ be a unit vector directed along the M -dimensional space. We will represent $\mathbf{u}_1$ as a D -component column vector.

It satisfies $\mathbf{u}_1^T \mathbf{u}_1 = 1$.

Projected position of data point $\mathbf{x}_n$ onto the space = $\mathbf{u}_1^T \mathbf{x}_n$.

Mean projected position of the data set = $\mathbf{u}_1^T \bar{\mathbf{x}}$ where

$$\bar{\mathbf{x}} = \frac{1}{N} \sum_{n=1}^N \mathbf{x}_n .$$



The Variance

The variance of the projected data is given by

$$\sigma^2 = \frac{1}{N} \sum_{n=1}^N [\mathbf{u}_1^T \mathbf{x}_n - \mathbf{u}_1^T \bar{\mathbf{x}}]^2 = \mathbf{u}_1^T \mathbf{S} \mathbf{u}_1,$$

where $\mathbf{S}$ is the covariance matrix given by

$$\mathbf{S} = \frac{1}{N} \sum_{n=1}^N (\mathbf{x}_n - \bar{\mathbf{x}})(\mathbf{x}_n - \bar{\mathbf{x}})^T.$$

Explanation in Component Form

Each data point is a D -dimensional vector. Each component of the data point is denoted as $x_{i\mu}$, where $i = 1, \dots, D$ and $\mu = 1, \dots, N$.

Similarly, the components of the unit vector are given by u_i , where $i = 1, \dots, D$.

It satisfies $\sum_{i=1}^D u_i^2 = 1$.

Projected position of the μ^{th} data point onto the space = $\sum_{i=1}^D u_i x_{i\mu}$.

Mean projected position of the data set

$$= \frac{1}{N} \sum_{\mu=1}^N \left(\sum_{i=1}^D u_i x_{i\mu} \right) = \sum_{i=1}^D u_i \left(\frac{1}{N} \sum_{\mu=1}^N x_{i\mu} \right) = \sum_{i=1}^D u_i \bar{x}_i, \text{ where } \bar{x}_i = \frac{1}{N} \sum_{\mu=1}^N x_{i\mu}.$$

Multiplications

The variance of the projected data is given by

$$\sigma^2 = \frac{1}{N} \sum_{\mu=1}^N \left(\sum_{i=1}^D u_i x_{i\mu} - \sum_{i=1}^D u_i \bar{x}_i \right)^2 = \frac{1}{N} \sum_{\mu=1}^N \left[\sum_{i=1}^D u_i (x_{i\mu} - \bar{x}_i) \right] \left[\sum_{j=1}^D u_j (x_{j\mu} - \bar{x}_j) \right].$$

Interchanging the order of summation, the variance becomes

$$\sigma^2 = \sum_{i,j=1}^D u_i \left[\frac{1}{N} \sum_{\mu=1}^N (x_{i\mu} - \bar{x}_i)(x_{j\mu} - \bar{x}_j) \right] u_j = \sum_{i,j=1}^D u_i S_{ij} u_j$$

where S_{ij} is a $D \times D$ **covariance** matrix given by

$$S_{ij} = \frac{1}{N} \sum_{\mu=1}^N (x_{i\mu} - \bar{x}_i)(x_{j\mu} - \bar{x}_j).$$

2.2 Maximizing the Projected Variance

We now maximize the projected variance $\mathbf{u}_1^T \mathbf{S} \mathbf{u}_1$ with respect to $\mathbf{u}_1$. This is equivalent to a rotation of the unit vector in the D -dimensional space. The maximization is constrained by the normalization condition $\mathbf{u}_1^T \mathbf{u}_1 = 1$.

Introducing the Lagrange multiplier λ_1 , we consider the unconstrained maximization of the Lagrangian

$$L = \mathbf{u}_1^T \mathbf{S} \mathbf{u}_1 + \lambda_1 (1 - \mathbf{u}_1^T \mathbf{u}_1).$$

By setting the derivative with respect to $\mathbf{u}_1$ equal to zero,

$$\mathbf{S} \mathbf{u}_1 = \lambda_1 \mathbf{u}_1,$$

which says that $\mathbf{u}_1$ is an eigenvector of $\mathbf{S}$.

First Principal Component

$$\mathbf{S}\mathbf{u}_1 = \lambda_1 \mathbf{u}_1.$$

Left-multiplying by $\mathbf{u}_1^T$ and making use of $\mathbf{u}_1^T \mathbf{u}_1 = 1$,

$$\mathbf{u}_1^T \mathbf{S}\mathbf{u}_1 = \lambda_1.$$

Hence the variance is maximum when we set $\mathbf{u}_1$ equal to the eigenvector having the **largest eigenvalue** λ_1 . This eigenvector is known as the first **principal component**.

Explanation in Component Form

Introducing the Lagrange multiplier λ_1 , we consider the unconstrained maximization of the Lagrangian

$$L = \sum_{i,j=1}^D u_i S_{ij} u_j + \lambda_1 \left(1 - \sum_{i=1}^D u_i^2 \right).$$

By setting the derivative with respect to u_i equal to zero,

$$\sum_{j=1}^D S_{ij} u_j = \lambda_1 u_i,$$

which says that $\mathbf{u}_1$ is an eigenvector of $\mathbf{S}$.

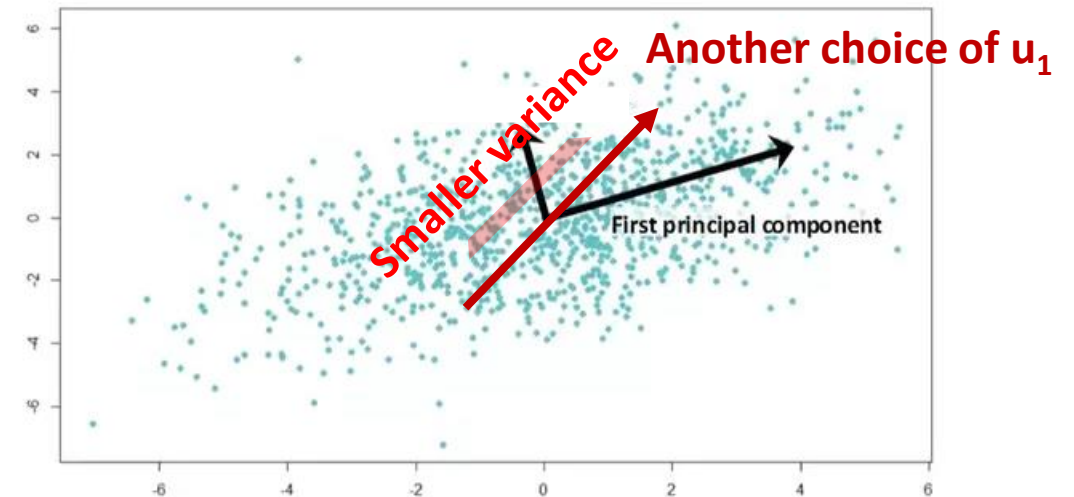
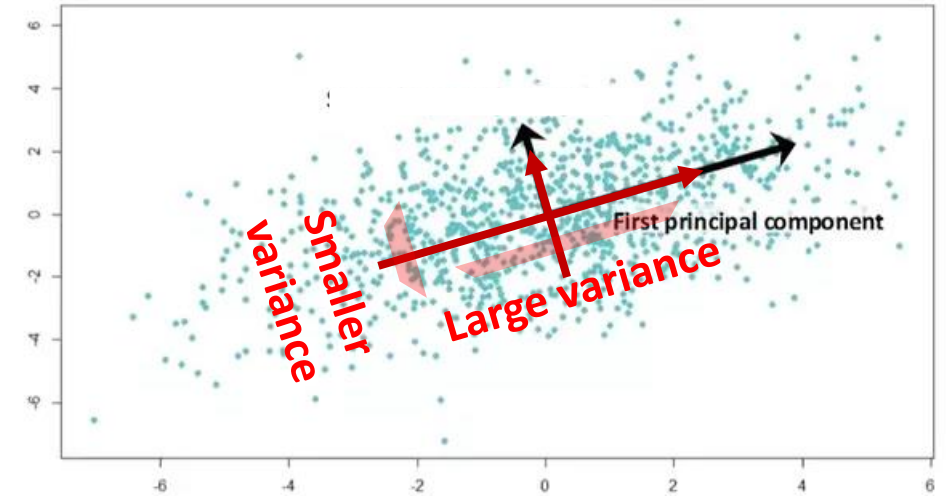
Multiplying both sides by u_i , summing over i , and making use of $\sum_{i=1}^D u_i^2 = 1$,

$$\sum_{i,j=1}^D u_i S_{ij} u_j = \lambda_1 \sum_{i=1}^D u_i^2 = \lambda_1.$$

2.3 Geometric Interpretation

In general, for projection onto an M -dimensional subspace, the variance is maximized when the data points are projected onto the space spanned by the M principal components, which are eigenvectors $\mathbf{u}_1, \dots, \mathbf{u}_M$ corresponding to the M largest eigenvalues $\lambda_1, \dots, \lambda_M$. This subspace is called the **principal subspace**.

We may interpret the projection of a data point onto this maximum variance subspace is **most descriptive** of the data point.



2.4 Orthonormality (1)

Since S_{ij} is a real symmetric matrix, the eigenvectors of different eigenvalues are orthogonal.

Proof: Let $\mathbf{u}_a$ and $\mathbf{u}_b$ be the eigenvectors corresponding to eigenvalues λ_a and λ_b respectively, and $\lambda_a \neq \lambda_b$. Then

$$\mathbf{S}\mathbf{u}_a = \lambda_a \mathbf{u}_a \quad \text{and} \quad \mathbf{S}\mathbf{u}_b = \lambda_b \mathbf{u}_b.$$

Left-multiplying by $\mathbf{u}_b^T$ and $\mathbf{u}_a^T$ respectively,

$$\mathbf{u}_b^T \mathbf{S} \mathbf{u}_a = \lambda_a \mathbf{u}_b^T \mathbf{u}_a \quad \text{and} \quad \mathbf{u}_a^T \mathbf{S} \mathbf{u}_b = \lambda_b \mathbf{u}_a^T \mathbf{u}_b.$$

Considering the transpose of the first equation,

$$\mathbf{u}_a^T \mathbf{S}^T \mathbf{u}_b = \lambda_a \mathbf{u}_a^T \mathbf{u}_b.$$

2.4 Orthonormality (2)

Since $\mathbf{S}$ is symmetric, $\mathbf{S}^T = \mathbf{S}$. Hence

$$\begin{aligned}\mathbf{u}_a^T \mathbf{S} \mathbf{u}_b &= \lambda_a \mathbf{u}_a^T \mathbf{u}_b = \lambda_b \mathbf{u}_a^T \mathbf{u}_b, \\ (\lambda_a - \lambda_b) \mathbf{u}_a^T \mathbf{u}_b &= 0.\end{aligned}$$

Since $\lambda_a \neq \lambda_b$, we arrive at the orthogonality condition:

$$\mathbf{u}_a^T \mathbf{u}_b = 0.$$

Requiring that the eigenvectors are normalized, $\mathbf{u}_a^T \mathbf{u}_a = 1$,

$$\mathbf{u}_a^T \mathbf{u}_b = \delta_{ab}.$$

We say that the eigenvectors are **orthonormal** (標準正交), that is, both orthogonal and normalized.

Proof in Component Form (1)

$$\sum_{j=1}^D S_{ij} u_j^a = \lambda_a u_i^a \quad \text{and} \quad \sum_{j=1}^D S_{ij} u_j^b = \lambda_b u_i^b.$$

Multiplying by u_i^b and u_i^a respectively, and summing over i ,

$$\sum_{i,j=1}^D u_i^b S_{ij} u_j^a = \lambda_a \sum_{i=1}^D u_i^b u_i^a \quad \text{and} \quad \sum_{i,j=1}^D u_i^a S_{ij} u_j^b = \lambda_b \sum_{i=1}^D u_i^a u_i^b.$$

Interchanging the dummy indices i and j in the first summation,

$$\sum_{i,j=1}^D u_j^b S_{ji} u_i^a = \lambda_a \sum_{i=1}^D u_i^a u_i^b,$$

Proof in Component Form (2)

Since $\mathbf{S}$ is symmetric, $S_{ij} = S_{ji}$. Hence

$$\sum_{i,j=1}^D u_i^a S_{ij} u_j^b = \lambda_a \sum_{i=1}^D u_i^a u_i^b = \lambda_b \sum_{i=1}^D u_i^a u_i^b ,$$
$$(\lambda_a - \lambda_b) \sum_{i=1}^D u_i^a u_i^b = 0.$$

Since $\lambda_a \neq \lambda_b$, we arrive at the orthogonality condition:

$$\sum_{i=1}^D u_i^a u_i^b = \delta_{ab}.$$

2.5 Completeness (1)

All vectors in the D -dimensional space can be expressed as a linear combination of the eigenvectors. We say that the eigenvectors form a complete set. Hence for all $\mathbf{x}$,

$$\mathbf{x} = \sum_{a=1}^D \alpha_a \mathbf{u}_a .$$

Left-multiplying by $\mathbf{u}_b^T$, and using the orthonormality condition,

$$\mathbf{u}_b^T \mathbf{x} = \sum_{a=1}^D \alpha_a \mathbf{u}_b^T \mathbf{u}_a = \sum_{a=1}^D \alpha_a \delta_{ba} = \alpha_b .$$

This implies

$$\mathbf{x} = \sum_{a=1}^D \mathbf{u}_a (\mathbf{u}_a^T \mathbf{x}) .$$

2.5 Completeness (2)

In component form,

$$x_i = \sum_{a=1}^D u_i^a \left(\sum_{j=1}^D u_j^a x_j \right).$$

Interchanging the order of summation,

$$x_i = \sum_{j=1}^D \left(\sum_{a=1}^D u_i^a u_j^a \right) x_j.$$

Since the vector $\mathbf{x}$ is arbitrary, this implies that

$$\sum_{a=1}^D u_i^a u_j^a = \delta_{ij}.$$

This is called the **completeness** (完整性) condition.

2.6 Unitary Matrix

Consider the matrix formed by pooling the eigenvectors together.

$$\mathbf{U} = (\mathbf{u}_1 \cdots \mathbf{u}_D).$$

Then the orthonormality of the eigenvectors implies that

$$\sum_{i=1}^D u_i^a u_i^b = \delta_{ab} \Rightarrow \mathbf{U}^T \mathbf{U} = \mathbf{I}.$$

The completeness of the eigenvectors implies that

$$\sum_{a=1}^D u_i^a u_j^a = \delta_{ij} \Rightarrow \mathbf{U} \mathbf{U}^T = \mathbf{I}.$$

This shows that $\mathbf{U}^{-1} = \mathbf{U}^T$ and $\mathbf{U} = (\mathbf{U}^T)^{-1}$. $\mathbf{U}$ is said to be **unitary** (酉).

2.7 Matrix Diagonalization

Writing the eigenvalues and eigenvectors of the covariance matrix in a single expression,

$$\mathbf{S}(\mathbf{u}_1 \cdots \mathbf{u}_D) = (\mathbf{u}_1 \cdots \mathbf{u}_D) \begin{pmatrix} \lambda_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_D \end{pmatrix}.$$

Introducing the $D \times D$ matrix $\mathbf{L} \equiv \begin{pmatrix} \lambda_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_D \end{pmatrix}$, we can write

$$\mathbf{S}\mathbf{U} = \mathbf{U}\mathbf{L}.$$

Note that the unitarity of $\mathbf{U}$ implies that

$$\mathbf{S} = \mathbf{U}\mathbf{L}\mathbf{U}^T \text{ and } \mathbf{L} = \mathbf{U}^T\mathbf{S}\mathbf{U}.$$

The process of transforming the matrix $\mathbf{S}$ to $\mathbf{L}$ is called **matrix diagonalization** (対角化).

Summary of Eigenvector Properties

Orthonormality: $\sum_{i=1}^D u_i^a u_i^b = \delta_{ab}$ or $\mathbf{U}^T \mathbf{U} = \mathbf{I}$.

Completeness: $\sum_{a=1}^D u_i^a u_j^a = \delta_{ij}$ or $\mathbf{U} \mathbf{U}^T = \mathbf{I}$.

Unitarity: $\mathbf{U} = (\mathbf{U}^T)^{-1}$

Matrix diagonalization: $\mathbf{S} = \mathbf{U} \mathbf{L} \mathbf{U}^T$.

Summary of the PCA

For a D -dimensional data set with N data points,

Calculate the mean $\bar{\mathbf{x}}$.

Calculate the covariance matrix $\mathbf{S}$.

Calculate the M largest eigenvalues and their corresponding eigenvectors.

The projection of a data point $\mathbf{x}$ onto the principal subspace is given by

$$\mathbf{x}_{PCA} = \bar{\mathbf{x}} + \sum_{i=1}^M \mathbf{u}_i [\mathbf{u}_i^T (\mathbf{x} - \bar{\mathbf{x}})],$$

where $\mathbf{u}_i$ are the normalized eigenvectors of λ_i .

3. Minimum Error Formulation

3.1 The Approximation

In this formulation, the goal is to minimize the projection error.

To do this, we introduce a complete orthonormal set of D -dimensional basis vectors $\{\mathbf{u}_a\}$ where $a = 1, \dots, D$ that satisfy

$$\mathbf{u}_a^T \mathbf{u}_b = \delta_{ab}.$$

Because this basis is complete, each data point can be represented exactly by a linear combination of the basis vectors

$$\mathbf{x}_n = \sum_{a=1}^D \alpha_{na} \mathbf{u}_a .$$

Rotation

This corresponds to a **rotation** of the coordinate system to a new system defined by $\{\mathbf{u}_a\}$, and the original D components $\{x_{n1}, \dots, x_{nD}\}$ are now replaced by an equivalent set $\{\alpha_{n1}, \dots, \alpha_{nD}\}$.

Taking the inner product with $\mathbf{u}_b$, and making use of the orthonormality property, we obtain

$$\mathbf{u}_b^T \mathbf{x}_n = \sum_{i=1}^D \alpha_{ni} \mathbf{u}_b^T \mathbf{u}_i = \sum_{i=1}^D \alpha_{ni} \delta_{bi} = \alpha_{nb}.$$

Therefore,

$$\mathbf{x}_n = \sum_{a=1}^D \mathbf{u}_a (\mathbf{u}_a^T \mathbf{x}_n).$$

The Approximation Goal

Our goal is to approximate this data point using a representation with a restricted number $M < D$ of variables corresponding to a projection onto a lower-dimensional subspace.

Consider the approximation:

$$\tilde{\mathbf{x}}_n = \sum_{a=1}^M z_{na} \mathbf{u}_a + \sum_{a=M+1}^D b_a \mathbf{u}_a .$$

3.2 The Distortion Measure (1)

Our goal is to minimize the distortion measure

$$J = \frac{1}{N} \sum_{n=1}^N \|\mathbf{x}_n - \tilde{\mathbf{x}}_n\|^2.$$

Minimizing with respect to z_{na} ($a = 1, \dots, M$),

$$\frac{\partial J}{\partial z_{na}} = \frac{2}{N} \sum_{i=1}^D (x_{ni} - \tilde{x}_{ni}) u_i^a = \frac{2}{N} (\mathbf{x}_n^T \mathbf{u}_a - \tilde{\mathbf{x}}_n^T \mathbf{u}_a) = 0.$$

Since

$$\tilde{\mathbf{x}}_n^T \mathbf{u}_a = \sum_{i=1}^M z_{ni} \mathbf{u}_i^T \mathbf{u}_a + \sum_{i=M+1}^D b_i \mathbf{u}_i^T \mathbf{u}_a = \sum_{i=1}^M z_{ni} \delta_{ia} + 0 = z_{na},$$

we have

$$z_{na} = \tilde{\mathbf{x}}_n^T \mathbf{u}_a = \mathbf{x}_n^T \mathbf{u}_a.$$

3.2 The Distortion Measure (2)

Minimizing with respect to b_a ($a = M + 1, \dots, D$),

$$\frac{\partial J}{\partial b_a} = \frac{2}{N} \sum_{n=1}^N (x_{ni} - \tilde{x}_{ni}) u_i^a = \frac{2}{N} \sum_{n=1}^N (\mathbf{x}_n^T \mathbf{u}_a - \tilde{\mathbf{x}}_n^T \mathbf{u}_a) = 0.$$

Since

$$\tilde{\mathbf{x}}_n^T \mathbf{u}_a = \sum_{i=1}^M z_{ni} \mathbf{u}_i^T \mathbf{u}_a + \sum_{i=M+1}^D b_i \mathbf{u}_i^T \mathbf{u}_a = 0 + \sum_{i=1}^M b_i \delta_{ia} = b_a,$$

we have

$$b_a = \frac{1}{N} \sum_{n=1}^N \mathbf{x}_n^T \mathbf{u}_a = \bar{\mathbf{x}}^T \mathbf{u}_a.$$

The Displacement Vector

Substituting these results into the expression of $\mathbf{x} - \tilde{\mathbf{x}}$,

$$\begin{aligned}\mathbf{x}_n - \tilde{\mathbf{x}}_n &= \sum_{a=1}^D (\mathbf{x}_n^T \mathbf{u}_a) \mathbf{u}_a - \sum_{a=1}^M (\mathbf{x}_n^T \mathbf{u}_a) \mathbf{u}_a - \sum_{a=M+1}^D (\bar{\mathbf{x}}^T \mathbf{u}_a) \mathbf{u}_a \\ &= \sum_{a=M+1}^D [(\mathbf{x}_n - \bar{\mathbf{x}})^T \mathbf{u}_a] \mathbf{u}_a.\end{aligned}$$

The displacement vector from $\mathbf{x}_n$ to $\tilde{\mathbf{x}}_n$ lies in the space orthogonal to the principal subspace, because it is a linear combination of $\{\mathbf{u}_a\}$ for $a = M + 1, \dots, D$.

Therefore, the expression of the distortion measure J becomes

$$J = \frac{1}{N} \sum_{n=1}^N \sum_{a=M+1}^D [(\mathbf{x}_n - \bar{\mathbf{x}})^T \mathbf{u}_a]^2.$$

In Terms of The Covariance Matrix

In component form,

$$J = \frac{1}{N} \sum_{n=1}^N \sum_{a=M+1}^D \left[\sum_{i=1}^D (x_{ni} - \bar{x}_{ni}) u_i^a \right]^2 .$$

Interchanging the order of summation,

$$J = \sum_{a=M+1}^D \sum_{i,j=1}^D u_i^a \left[\frac{1}{N} \sum_{n=1}^N (x_{ni} - \bar{x}_{ni})(x_{nj} - \bar{x}_{nj}) \right] u_j^a .$$

In matrix form involving the covariance matrix,

$$J = \sum_{a=M+1}^D \mathbf{u}_a^T \mathbf{S} \mathbf{u}_a .$$

Minimizing the Distortion Measure

Minimizing J with the constraints $\mathbf{u}_a^T \mathbf{u}_a = 1$ for $a = M + 1, \dots, D$, we introduce the Lagrange multipliers λ_a , and consider the unconstrained maximization of the Lagrangian

$$L = \sum_{a=M+1}^D [\mathbf{u}_a^T \mathbf{S} \mathbf{u}_a + \lambda_a (1 - \mathbf{u}_a^T \mathbf{u}_a)].$$

By setting the derivative with respect to $\mathbf{u}_a$ equal to zero,

$$\mathbf{S} \mathbf{u}_a = \lambda_a \mathbf{u}_a.$$

Left-multiplying by $\mathbf{u}_a^T$ and making use of $\mathbf{u}_a^T \mathbf{u}_a = 1$,

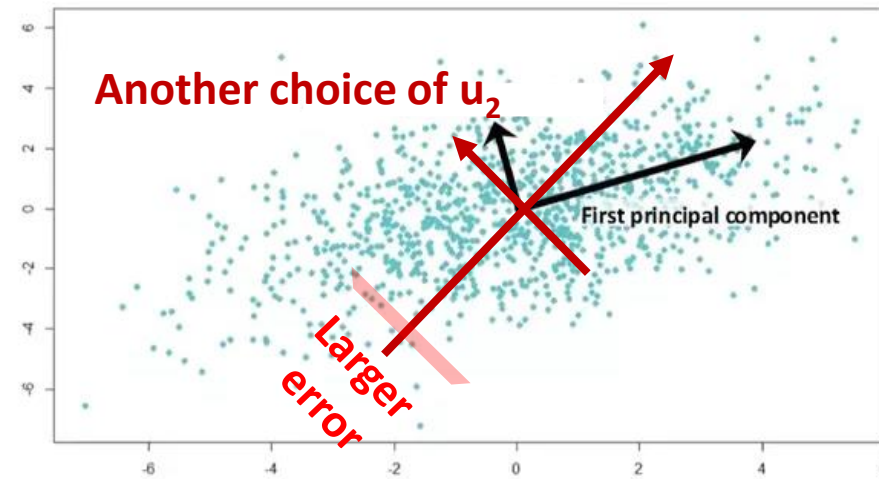
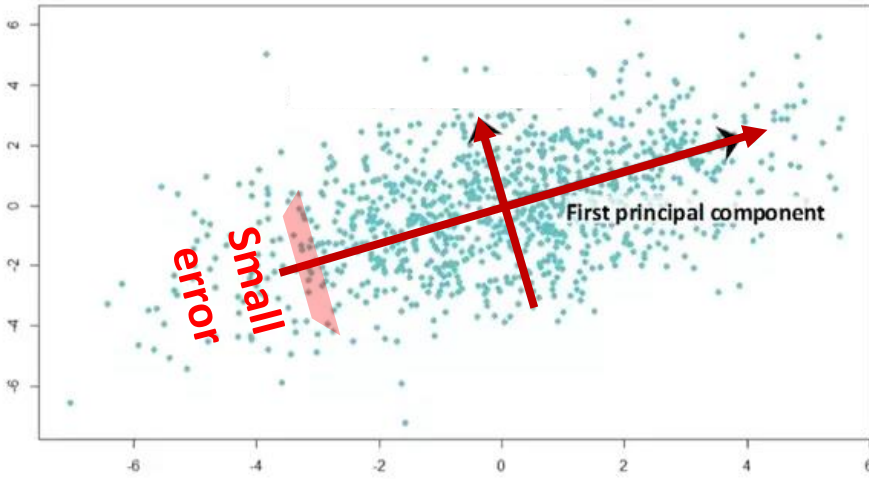
$$J = \sum_{a=M+1}^D \mathbf{u}_a^T \mathbf{S} \mathbf{u}_a = \sum_{a=M+1}^D \lambda_a.$$

The Result

$$J = \sum_{a=M+1}^D \lambda_a .$$

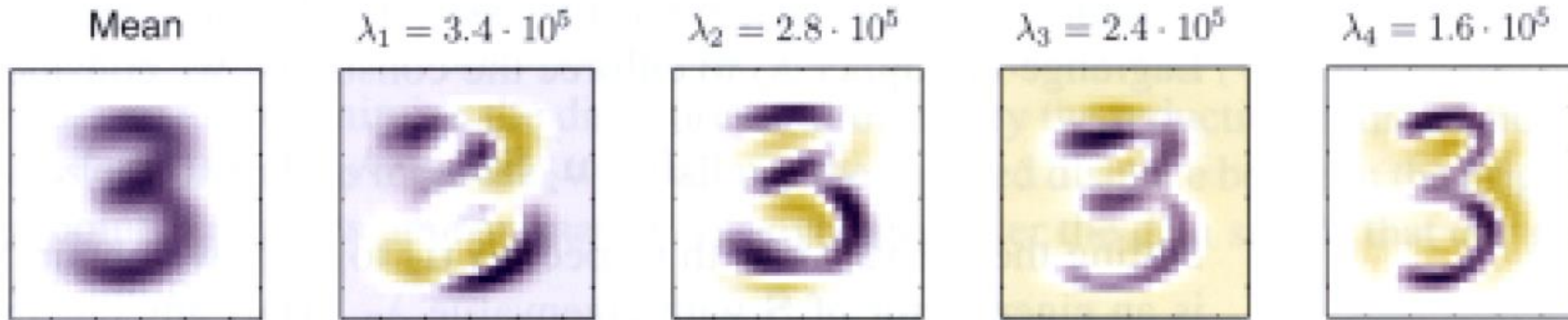
Hence the distortion measure is minimum when we set $\mathbf{u}_a$ equal to the eigenvectors having the **smallest $D - M$ eigenvalues** λ_a . The eigenvalues defining the principal subspace have the **largest M eigenvalues**.

3.3 Geometric Interpretation



4. Applications of PCA

4.1 Data Compression



The mean vector $\bar{x}$ along with the first four PCA eigenvectors $u_1, \dots, u_4$ for the off-line digits data set, together with the corresponding eigenvalues.

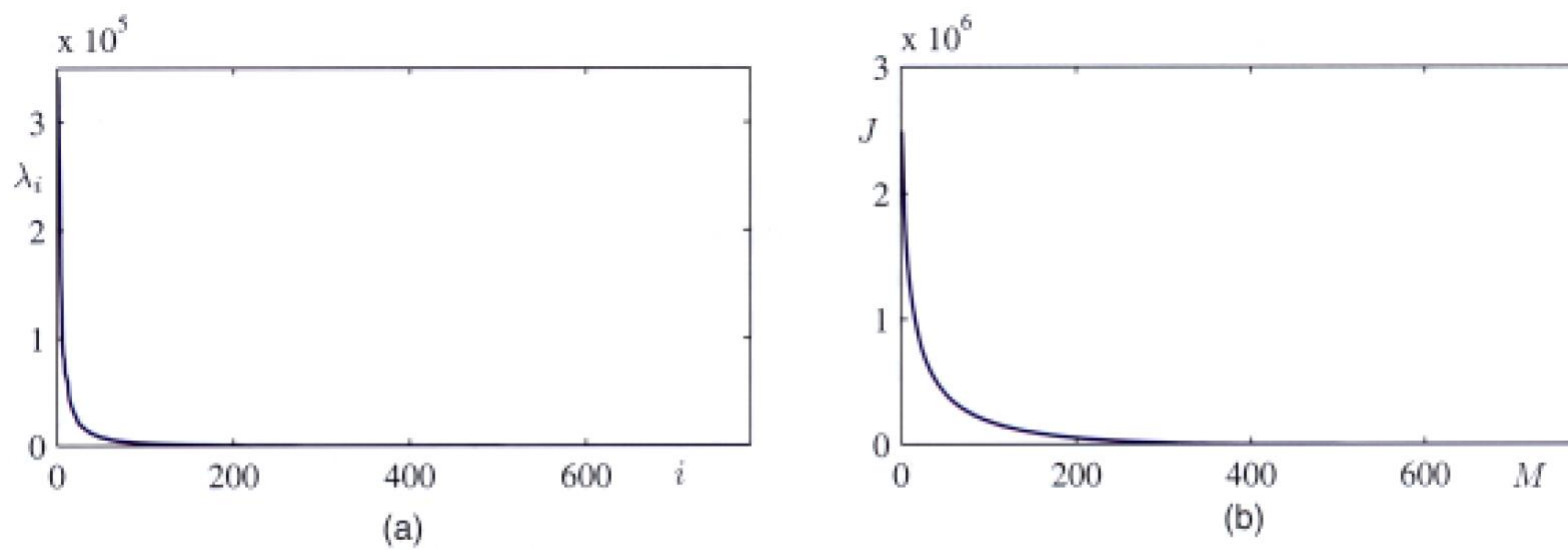
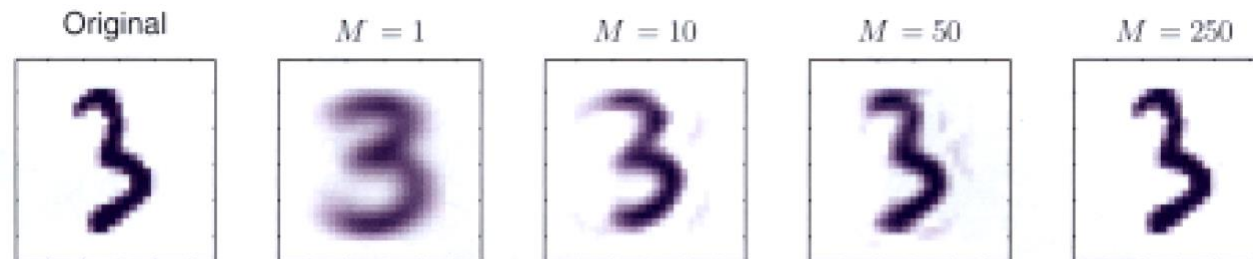


Figure 12.4 (a) Plot of the eigenvalue spectrum for the off-line digits data set. (b) Plot of the sum of the discarded eigenvalues, which represents the sum-of-squares distortion J introduced by projecting the data onto a principal component subspace of dimensionality M .

$$\text{Reconstruction: } \tilde{\mathbf{x}} = \bar{\mathbf{x}} + \sum_{i=1}^M \mathbf{u}_i [\mathbf{u}_i^T (\mathbf{x} - \bar{\mathbf{x}})],$$



An original example from the off-line digits data set together with its PCA reconstructions obtained by retaining M principal components for various values of M . As M increases the reconstruction becomes more accurate and would become perfect when $M = D = 28 \times 28 = 784$.

4.2 Data Pre-processing

The goal is not dimensionality reduction but rather the transformation of a data set in order to standardize certain of its properties.

For data with inputs with a wide range of magnitudes, a common way to standardize the data is by using the **correlation matrix**,

$$\rho_{ij} = \frac{1}{N} \sum_{n=1}^N \left(\frac{x_{ni} - \bar{x}_i}{\sigma_i} \right) \left(\frac{x_{nj} - \bar{x}_j}{\sigma_j} \right).$$

PCA offers another way to standardize the data.

Then we define, for each data point $\mathbf{x}_n$, a transformed value given by

$$\mathbf{y}_n = \mathbf{L}^{-1/2} \mathbf{U}^T (\mathbf{x}_n - \bar{\mathbf{x}}).$$

Here $\mathbf{L}^{-1/2} = \begin{pmatrix} \lambda_1^{-1/2} & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \lambda_N^{-1/2} \end{pmatrix}$. It is easily seen that $\mathbf{y}_n$ has a zero mean.

Data Whitening

For the covariance,

$$\begin{aligned}\frac{1}{N} \sum_{n=1}^N \mathbf{y}_n \mathbf{y}_n^T &= \frac{1}{N} \sum_{n=1}^N \mathbf{L}^{-1/2} \mathbf{U}^T (\mathbf{x}_n - \bar{\mathbf{x}}) (\mathbf{x}_n - \bar{\mathbf{x}})^T \mathbf{U} \mathbf{L}^{-1/2} . \\ &= \mathbf{L}^{-1/2} \mathbf{U}^T \mathbf{S} \mathbf{U} \mathbf{L}^{-1/2} = \mathbf{L}^{-1/2} \mathbf{L} \mathbf{L}^{-1/2} = \mathbf{I} .\end{aligned}$$

This operation is known as **whitening** or **sphereing** the data.

Google: Data preprocessing: whitening or sphereing in python

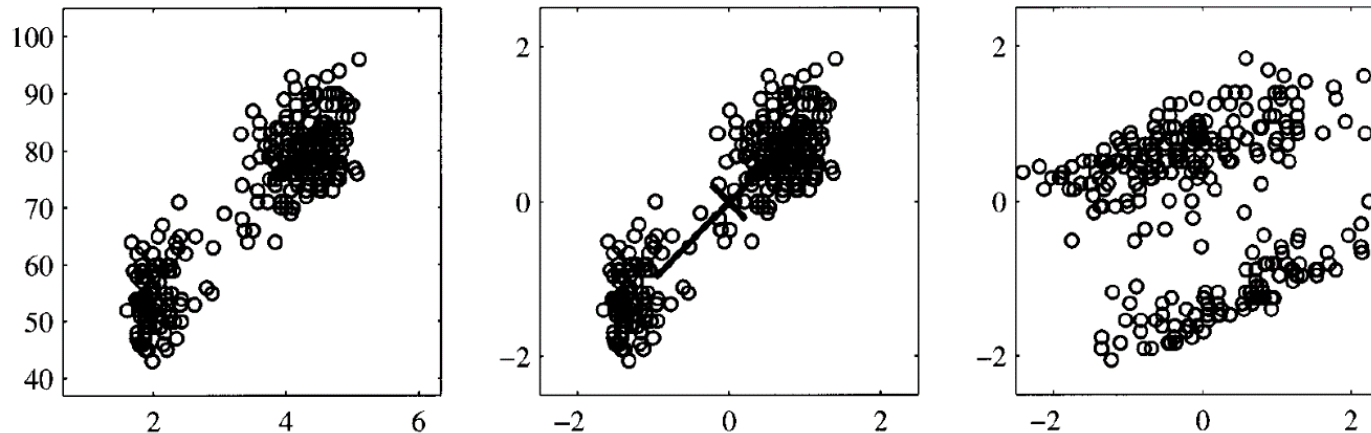
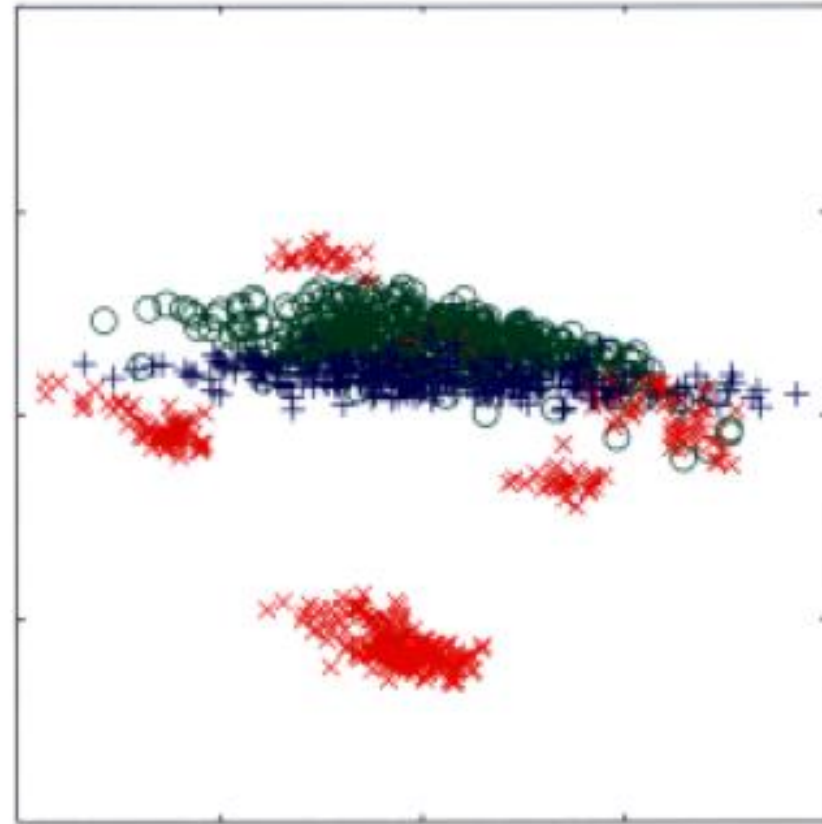


Figure 12.6 Illustration of the effects of linear pre-processing applied to the Old Faithful data set. The plot on the left shows the original data. The centre plot shows the result of standardizing the individual variables to zero mean and unit variance. Also shown are the principal axes of this normalized data set, plotted over the range $\pm \lambda_i^{1/2}$. The plot on the right shows the result of whitening of the data to give it zero mean and unit covariance.

4.3 Data Visualization

Each data point is projected onto a two-dimensional ($M = 2$) principal subspace. A data point $\mathbf{x}_n$ is plotted at Cartesian coordinates given by $\mathbf{x}_n^T \mathbf{u}_1$ and $\mathbf{x}_n^T \mathbf{u}_2$.

Visualization of the oil flow data set obtained by projecting the data onto the first two principal components. The red, blue, and green points correspond to the 'laminar', 'homogeneous', and 'annular' flow configurations respectively.



5. Application to Portfolio Management in Finance³

5.1 Principal Portfolios

- To reduce risks, investments are organized in **portfolios** consisting of a collection of securities.
- **Diversification of risks** becomes important after the 2008 financial crisis.
- Securities that have **low correlations** or even negative correlations could eliminate some risk.
- Consider the **covariance or correlation matrix** of the returns of a set of securities.
- The **eigenvectors** of PCA describe how a number of security returns are correlated.
- When the prices are driven by external inputs, the prices of the securities belonging to an eigenvector **move together**.
- Due to the orthogonality of eigenvectors, the price movements of different eigenvectors are **independent**, each being driven a particular source of risk.
- Investments according to the PCA eigenvectors are called **principal portfolios**.

³ Yang, Libin (2015) *An Application of Principal Component Analysis to Stock Portfolio Management*.

5.2 Risk Diversification of Stocks

(A) Stock Investment with Equal Weight

In this strategy, the **capital allocated to each stock** in the portfolio **is the same**.

(B) Risk Parity (or Equal Risk Contribution)

In this strategy, the risk of stock i per unit of capital is estimated by its standard deviation σ_i (note that the risk correlations of the stocks are neglected). The capital allocated to each stock in the portfolio is such that **the risk of each stock is the same**. Hence the capital allocated to each stock is proportional to $1/\sigma_i$.

Risk Diversification of Portfolios

(C) Equal Weight Portfolio (or 1/N Strategy)

In this strategy, the first N principal portfolios are selected, and **equal weight is allocated to each principal portfolio**.

Since eigenvectors remain to be eigenvectors when the signs are flipped, the first step is to determine the signs of the eigenvectors $\pm \mathbf{u}_a$ to be adopted.

For example, we may fix the sign by requiring the projection of this portfolio to the portfolio with equal weight of stocks to be positive.

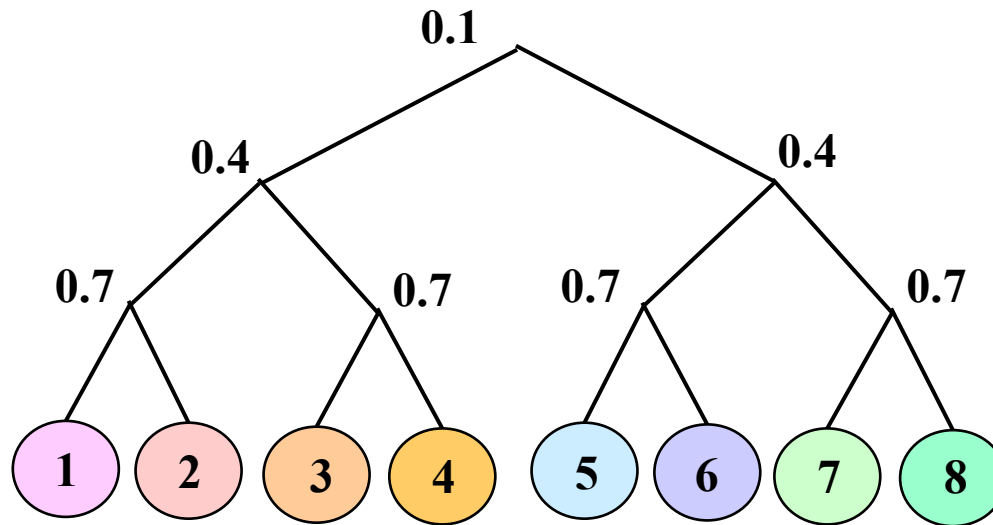
(D) Diversified Risk Parity

In this strategy, the capital allocated to each of the N principal portfolios are such that **the risk of each portfolio is the same**. Since the eigenvalue of each portfolio is the variance λ_a , the capital allocated to each portfolio is proportional to $1/\sqrt{\lambda_a}$.

5.3 PCA of a Hierarchically Grouped Matrix

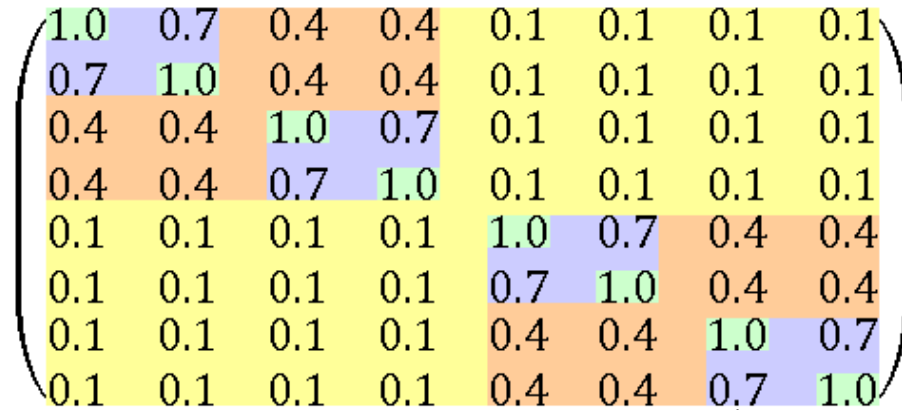
Consider a stock market in which the stocks are **hierarchically organized**. This means that the stocks are divided into groups, and groups are composed of subgroups, and so on. In investment management, it is common to use the **correlation matrix** to avoid domination by stocks with high variance.

Here we consider an example of 8 stocks related by:



The Correlation Matrix

The correlation matrix is given by



1.0	0.7	0.4	0.4	0.1	0.1	0.1	0.1
0.7	1.0	0.4	0.4	0.1	0.1	0.1	0.1
0.4	0.4	1.0	0.7	0.1	0.1	0.1	0.1
0.4	0.4	0.7	1.0	0.1	0.1	0.1	0.1
0.1	0.1	0.1	0.1	1.0	0.7	0.4	0.4
0.1	0.1	0.1	0.1	0.7	1.0	0.4	0.4
0.1	0.1	0.1	0.1	0.4	0.4	1.0	0.7
0.1	0.1	0.1	0.1	0.4	0.4	0.7	1.0

The eigenvalues and eigenvectors are given by

Eigenvalue	Eigenvector
2.9	$(1, 1, 1, 1, 1, 1, 1, 1)^T / \sqrt{8}$
2.1	$(1, 1, 1, 1, -1, -1, -1, -1)^T / \sqrt{8}$
0.9	$(1, 1, -1, -1, 0, 0, 0, 0)^T / 2$
0.9	$(0, 0, 0, 0, 1, 1, -1, -1)^T / 2$
0.3	$(1, -1, 0, 0, 0, 0, 0, 0)^T / \sqrt{2}$
0.3	$(0, 0, 1, -1, 0, 0, 0, 0)^T / \sqrt{2}$
0.3	$(0, 0, 0, 0, 1, -1, 0, 0)^T / \sqrt{2}$
0.3	$(0, 0, 0, 0, 0, 0, 1, -1)^T / \sqrt{2}$

Observations

- The first principal component corresponds to marketwide effects.
- The next principal component corresponds to a group of stocks.
- The next-next principal components correspond to subgroups, and so on.
- The smallest principal components may be due to random fluctuations.

Eigenvalue	Eigenvector
2.9	$(1, 1, 1, 1, 1, 1, 1, 1)^T / \sqrt{8}$
2.1	$(1, 1, 1, 1, -1, -1, -1, -1)^T / \sqrt{8}$
0.9	$(1, 1, -1, -1, 0, 0, 0, 0)^T / 2$
0.9	$(0, 0, 0, 0, 1, 1, -1, -1)^T / 2$
0.3	$(1, -1, 0, 0, 0, 0, 0, 0)^T / \sqrt{2}$
0.3	$(0, 0, 1, -1, 0, 0, 0, 0)^T / \sqrt{2}$
0.3	$(0, 0, 0, 0, 1, -1, 0, 0)^T / \sqrt{2}$
0.3	$(0, 0, 0, 0, 0, 0, 1, -1)^T / \sqrt{2}$

Remarks

- Both $\mathbf{u}_a$ and $-\mathbf{u}_a$ are eigenvectors. PCA does not imply the preference of long or short sale.
- When some eigenvalues are the same, the linear combinations of their eigenvectors are also eigenvectors.
- In practice, all $\pm$ sign combinations should be considered.
- The choice of the $\pm$ sign combination may be determined by the initial investment or by further comparison with real data.
- PCA is applicable when short sale is allowed. (Quadratic optimization is required when short sale is not allowed.)

5.4 How Many Principal Components Should be Included?

Some empirical rules:

(1) Cumulative variance: $V_m = \sum_{k=1}^m \lambda_k / p$ (p = the sum of all variances = total number of variables). When V_m exceeds 70% or 90%, m is cut off.

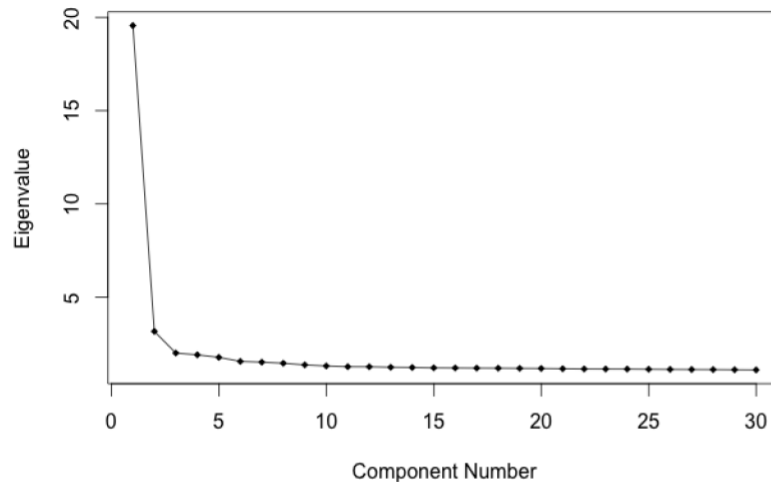
(2) Kaiser's rule: Only principal components with eigenvalues > 1 are retained. (If all stocks are uncorrelated, all eigenvalues become 1. So, according to this rule, any principal components with eigenvalues < 1 are not worth retaining.)

(3) Diagram of eigenvalue or log-eigenvalue versus component number: Find the point where the linear fits to the steep and flat parts of the curve meet. This is extensively used in practice.

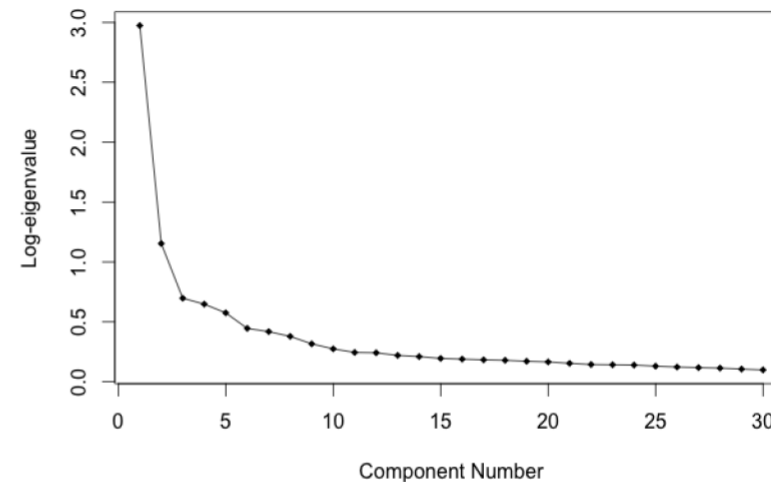
Diagram of Eigenvalue and Its Log

The author chose 10 principal components based on the plots.

Remark: A validation process should be more reliable.



(a) Scree graph for correlation matrix of 156 stocks, for the complete study period from 04/04/2000 to 17/02/2014.



(b) Log-eigenvalue (or LEV) diagram for correlation matrix of 156 stocks, for the complete study period from 04/04/2000 to 17/02/2014.

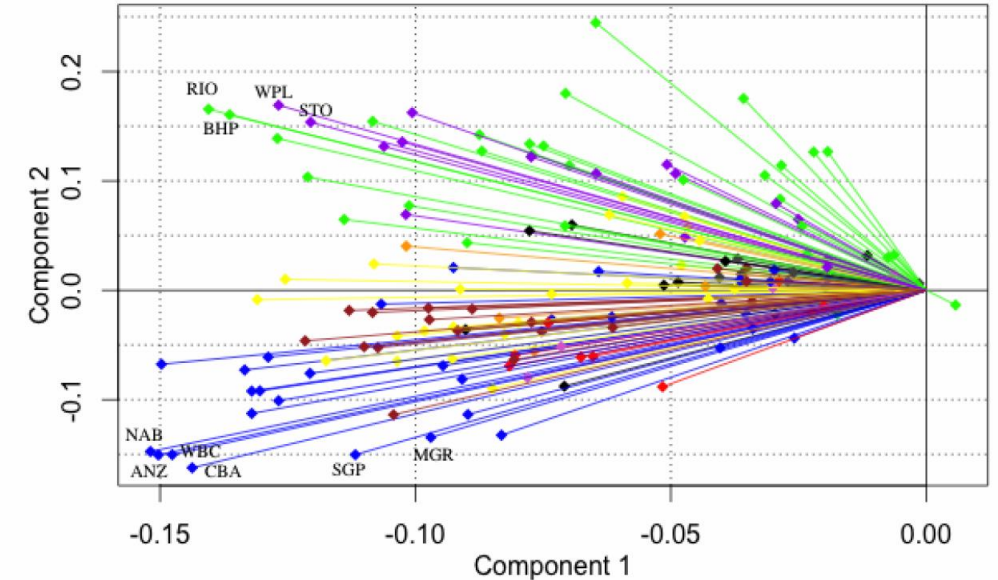
Visualization (1)

Component 1: All stocks have the same sign.

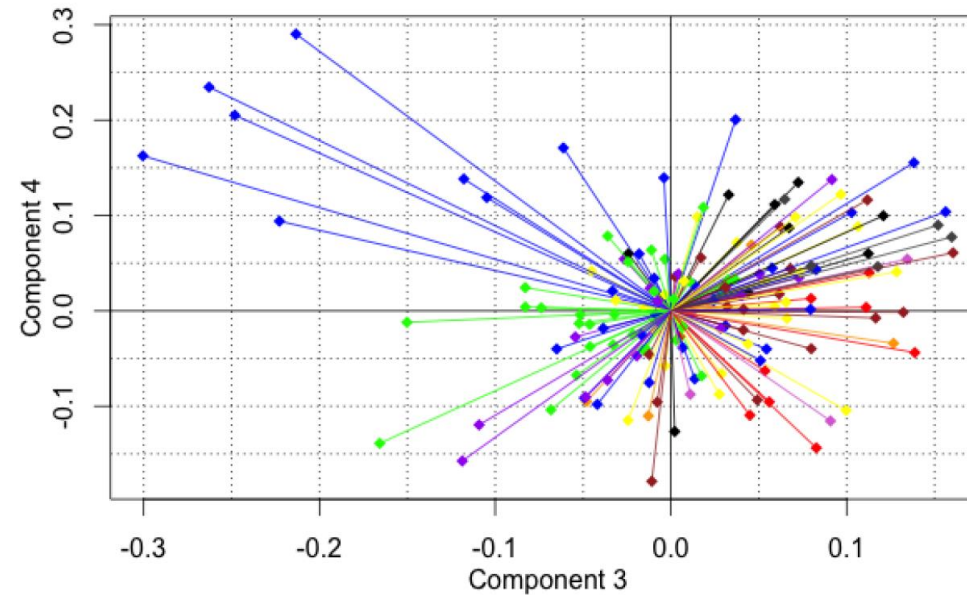
Component 2:

Positive groups of basic materials (green), oil & gas (purple), and consumer goods (black).

Negative groups of financials (blue), health care (red), and consumer services (brown). Neutral group of industrials (yellow).



Visualization (2)



Component 3: Positive groups of health care (red), consumer services (brown) and industrials (yellow).

Demonstration

See the files PCA_HSI.docx and PCA_demo_HSI.xlsm.